

Signal Generators from some manufacturers

Specifications	Tedyne LeCroy		Scientific India	Sciencetech Technologies Pvt. Ltd.	
	WaveStation 2012	WaveStation 2022	DG1022	Model Sciencetech 4210 1mHz-10MHz Function/Arbitrary Waveform Generator	
Waveforms	Sine, square, ramp, pulse, noise, arbitrary; stairup, staidown, positive pulse, negative pulse, up ramp, down ramp, sinc, Gaussian, LogFall, LogRise, Sqrt, TwoTone, etc	Sine, square, ramp, pulse, noise, arbitrary; stairup, staidown, positive pulse, negative pulse, up ramp, down ramp, sinc, Gaussian, LogFall, LogRise, Sqrt, TwoTone, etc	Sine, square, ramp	Sine, square, triangle, ramp, pulse and more than 25 arbitrary waveforms	
Frequency	1 μ Hz - 10 MHz	1 μ Hz - 25 MHz	20 MHz	1 mHz to 10 MHz	
Amplitude	2 mVpp - 20Vpp (depending on the CH and impedance)	2 mVpp - 20Vpp (depending on the CH and impedance)	2 mVpp - 10Vpp (CH1)	20Vpp (0.C), 10Vpp (50 Ω)	
Distortion	<0.2%	<0.2%	Harmonic distortion DC-20 kHz, 1Vpp<0.2%	<0.3 %	
Attenuation	NA	NA		Digitally synthesised amplitude, no external attenuation required	
Output impedance	50 Ω	50 Ω	50 Ω	50 Ω	
Rise/fall time	<12 ns	<12 ns	35 ns	<20 ns	
Sample rate	125 MSa/s	125 MSa/s	100 MSa/s	100 MSa/s	
Number of points in waveform	16 kpts/CH	16 kpts/CH		4096	
DC offset	± 1.5 V (50 Ω), ± 3 V (High Z) on CH1, ± 5 V (50 Ω), ± 10 V (High Z) on CH2	± 1.5 V (50 Ω), ± 3 V (High Z) on CH1, ± 5 V (50 Ω), ± 10 V (High Z) on CH2	± 1.5 V	± 9.5 V at amplitude 1Vpp (amplitude + DC offset = 20 Vpp max.)	
Floating output	NA	NA		Yes	
Sweep	Sine, square, ramp, arbitrary (1 ms-500 seconds)	Sine, Square, Ramp, Arbitrary (1 ms-500 seconds)	100 Hz $\leq$ Span $\leq$ 2 GHz, 10 ms to 2000 s	Linear sweep with user-defined start frequency, stop frequency and time of sweep	
Interfaces	USB and GPIB	USB and GPIB	LAN, USB host, USB device, VGA , GPIB (option)	RS232	
Frequency counter accuracy	± 50 ppm	± 50 ppm		NA	
Display	8.99cm (3.5 inch) TFT LCD	8.99cm (3.5 inch)TFT LCD	21.59cm (8.5-inch) widescreen LCD with 800x 80 resolution and fine spectral lines for displaying large data sets (601 display points in normal and 751 points in full-screen mode)	128x64 dot graphic monochrome LCD	
Dimensions	10.4x22.86x20.19 cm ³	10.4x22.86x20.19 cm ³	39.9x22.3x15.9 cm ³ , approximate	26.2x13x31.6 cm ³	
Additional features	>40 built-in arbitrary waveforms, graphical waveform editing software for PC	>40 built-in arbitrary waveforms, graphical waveform editing software for PC	3GHz tracking generator (DSA1030-TG)	Based on Direct Digital Synthesis (DDS) technique, pulse duty cycle from 5% to 95%, internal modulation-AM (variable depth of modulation), FM (variable deviation frequency), ASK/burst (variable number of cycle and time), FSK (variable F1, F2 and hop frequency) waveform creator software for custom waveform generation, non-volatile storage & recall for control panel setting	